

Rafael Mouallem Rosa

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Current Research

My current research develops the Sources of Uncertainty framework for Bayesian inference and explores its applications, with a focus on sparse Bayesian learning and Bayesian neural networks.

Academic Employment

Inria, Statify team <i>Postdoctoral Researcher</i>	Grenoble, France 2024–present
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Education

Ph.D. in Economics , São Paulo School of Economics, Fundação Getulio Vargas Advisor: Victor Filipe Martins-da-Rocha	2015–2019
Visiting Scholar , Harvard University, Department of Economics Host: Tomasz Strzalecki	2018–2019
M.S. in Economics , São Paulo School of Economics, Fundação Getulio Vargas	2013–2015
B.S. in Economics , PUC-SP	2005–2010

Publications and Manuscripts

Statistics and Machine Learning

- **Rosa, R.M.**, Arbel, J., and Nguyen, H.D. (2026). *Bayesian inference with sources of uncertainty: from confidence modelling to sparse estimation*. arXiv:2605.03134.
- **Rosa, R.M.**, Arbel, J., and Nguyen, H.D. (2026). *Bayesian inference with sources of uncertainty: a confidence-weighted approach to sparsity*. Accepted workshop paper, OPTIMAL: Optimisation and Post-Bayesian Inference in Machine Learning, AISTATS 2026 Workshop.
- **Rosa, R.M.** and Arbel, J. (2026). *Bayesian Structured Sparsity for Compact Architecture Recovery*. Manuscript.

Economics and Decision Theory

- Martins-da-Rocha, V.F. and **Rosa, R.M.** (2023). *Complete markets with bankruptcy risk and pecuniary default punishments*. *Economic Theory*, 75(3), 625–640.
- Martins-da-Rocha, V.F. and **Rosa, R.M.** (2020). *Second-Order Beliefs and Second-Order Expected Utility*. HAL preprint.
- Teles, V.K., Júnior, C.J., and **Rosa, R.M.** (2015). *Investment-specific technological change and the Brazilian macroeconomy*. *Brazilian Review of Econometrics*, 35(2), 233–264.

Industry Experience

Safra Asset Management <i>Quantitative Portfolio Manager</i>	São Paulo, Brazil 2020–2022
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- Designed and implemented quantitative trading strategies in futures and equities.
- Built research-to-production workflows for signal construction, backtesting, and risk monitoring.

Macro Capital Asset Management <i>Partner and Economist</i>	São Paulo, Brazil 2019–2020
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- Developed macro-finance quantitative models for investment decisions and scenario analysis.

Brasil Capital Asset Management <i>Analyst</i>	São Paulo, Brazil 2010–2011
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Votorantim Bank <i>Intern</i>	São Paulo, Brazil 2009–2010
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Teaching

São Paulo School of Economics, Fundação Getulio Vargas <i>Instructor, Probability, undergraduate course</i>	São Paulo, Brazil 2016
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Professional Service

Referee: *Bayesian Analysis*.

Awards and Scholarships

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| • CAPES PDSE Scholarship, Doctorate Abroad, Harvard University | 2018–2019 |
| • CAPES Scholarship, Ph.D. in Economics, FGV | 2015–2019 |
| • VII INFI-FEBRABAN Award, 3rd place, Category A | 2015 |
| • CAPES Scholarship, M.S. in Economics, FGV | 2013–2014 |

Technical Skills

- Python, R, MATLAB, $\text{\LaTeX}$